

Gradient function graphs



- 1
 - a 1
 - b 1
 - c A linear function has a constant gradient.

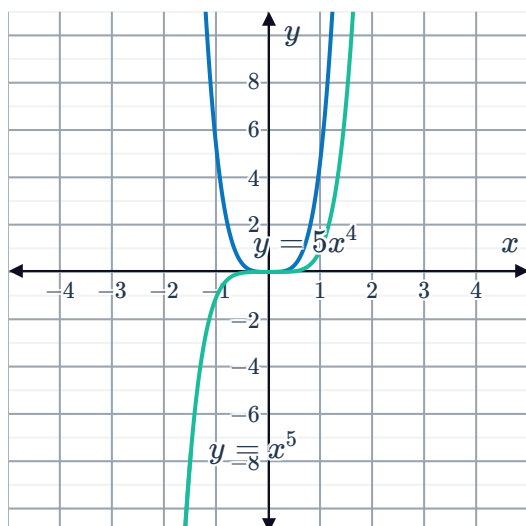
- 2
 - a The gradient function is positive when $y = x^2$ is increasing, and negative when $y = x^2$ is decreasing.
 - b Positive
 - c The gradient of the tangent line increases at a constant rate.
 - d Negative
 - e The gradient of the tangent line increases at a constant rate.
 - f Linear

- 3
 - a Positive
 - b The gradient of the tangent line increases at an increasing rate.
 - c Positive
 - d The gradient of the tangent line decreases at an decreasing rate.
 - e Quadratic

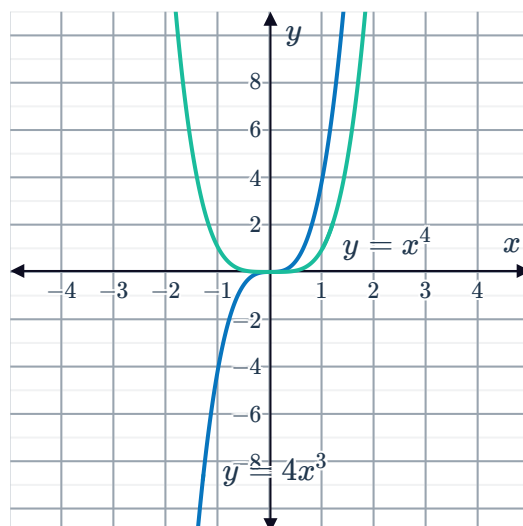
- 4
 - a Positive
 - b The gradient of the tangent line increases at an increasing rate.
 - c Negative
 - d The gradient of the tangent line increases at a decreasing rate.
 - e Cubic

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a



b



c Odd

d Even

Algebraic differentiation

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a

i Degree 1

ii Degree 2

iii Degree 3

b $n - 1$

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a

$$\frac{dy}{dx} = 7x^6$$

b

$$\frac{dy}{dx} = 9x^8$$

c

$$\frac{dy}{dx} = 5x^4$$

d

$$\frac{dy}{dx} = 0$$

Applications

8

$$f'(2) = 32$$

9

a 9

$$b \quad y = 9x - 8$$

10



a

Graph	y -intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	2
$y = x^3$	$(0, -2)$	3
$y = x^4$	$(0, -3)$	4
$y = x^5$	$(0, -4)$	5

b n c No, because then the gradient of the tangent at $x = 1$ would be 1.

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a False

b True

c False

d True